Matrix Group

First, we give some notes. We use $\mathbb{K}$ to represent the field $\mathbb{R}$ or $\mathbb{C}$. So we have

$$\mathrm{Mat}(n, \mathbb{K}) = \{n \times n \text{ matrix over } \mathbb{K}\} \tag{2.1}$$

which is not a group, due to A^{-1} doesn't have to exist for $A \in \mathrm{Mat}(n, \mathbb{K})$. But we can define

$$GL(n, \mathbb{K}) := \{A \in \mathrm{Mat}(n, \mathbb{K}) \mid \det A \neq 0\} \tag{2.2}$$

is a group: $\forall A, B \in GL(n, \mathbb{K})$, $AB \in GL(n, \mathbb{K})$ due to

$$\det(AB) = \det A \det B \neq 0. \tag{2.3}$$

and $\mathbb{I} \in GL(n, \mathbb{K})$, $A^{-1} \in GL(n, \mathbb{K})$. So it calls the general linear group of degree n over $\mathbb{K}$. We can also define the special linear group of degree n over $\mathbb{K}$ as

$$SL(n, \mathbb{K}) := \{A \in GL(n, \mathbb{K}) \mid \det A = 1\}. \tag{2.4}$$

($\det AB = \det A \det B = 1$), Thus,

$$\begin{array}{ccc} SL(n, \mathbb{C}) & < & GL(n, \mathbb{C}) \\ \vee & & \vee \\ SL(n, \mathbb{R}) & < & GL(n, \mathbb{R}) \end{array} \tag{2.5}$$

This drives us to a theorem:

定理 0..1 Theorem 7

$$SL(n, \mathbb{K}) \trianglelefteq GL(n, \mathbb{K}) \tag{2.6}$$

证明

$SL(n, \mathbb{K}) < GL(n, \mathbb{K})$. We can find the determinate is a homomorphism map:

$$\det = GL(n, \mathbb{K}) \to \mathbb{K}^* := \mathbb{K} \setminus \{0\}. \tag{2.7}$$

Since $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, then $\mathbb{K}^*$ is also a group. Obviously, $\ker(\det) = SL(n, \mathbb{K})$, thus base on the theorem 4: for homomorphism $\varphi : G \to G'$

- ① $\ker\varphi \trianglelefteq G$.
- ② $N \trianglelefteq G, \varphi(N) \trianglelefteq \varphi(G) < G'$

$$\Rightarrow SL(n, \mathbb{K}) \trianglelefteq GL(n, \mathbb{K}) \tag{2.8}$$

Base on the first isomorphism theorem:

$$GL(n, \mathbb{K})/\ker(\det) \cong \mathrm{Im}(\det) \tag{2.9}$$

$$\Rightarrow GL(n,\mathbb{K})/SL(n,\mathbb{K}) \cong \mathbb{K}^* \tag{2.10}$$

$$\Rightarrow \{A \cdot SL(n,\mathbb{K}) \mid A \in GL(n,\mathbb{K})\} \cong \mathbb{K}^* \tag{2.11}$$

We find that we can choose $\mathrm{diag}(1,1,\ldots,\lambda) \in GL(n,\mathbb{K})$ such that $\det(\mathrm{diag}(1,1,\ldots,\lambda)) = \lambda \in \mathbb{K}^*$. So the map det is onto map.

Now we give some definitions of useful groups.

定义 0..1 – Orthogonal group

$$O(n) = \{A \in GL(n,\mathbb{R}) \mid AA^T = \mathbb{I}\} \text{ with degree } n. \tag{2.12}$$

定义 0..2 – Unitary group

$$U(n) = \{A \in GL(n,\mathbb{C}) \mid AA^\dagger = \mathbb{I}\} \text{ with degree } n. \tag{2.13}$$

定义 0..3 – Symplectic group

$$Sp(2n) = \{A \in GL(2n,\mathbb{R}) \mid A^T J A = J\} \text{ with degree } 2n. \tag{2.14}$$

where J is a fixed, non-degenerate ($\det J \neq 0$) and skew-symmetric ($J^T = -J$) matrix.

There are some useful subgroups from $O(n)$ and $U(n)$:

- $SO(n) = \{A \in O(n) \mid \det A = 1\}$. (special orthogonal group) For $A \in O(n)$, $\det\big(AA^T\big) = \det A \det A^T = (\det A)^2 = 1$

$$\Rightarrow \det A = \pm 1. \tag{2.15}$$

$$\Rightarrow \det : O(n) \to Z_2 = \{+1,-1\} \text{ is a homomorphism} \tag{2.16}$$

Theorem 4: $\ker(\det) = SO(n) \trianglelefteq O(n)$,
First isomorphism theorem: $G/\ker\varphi \cong \mathrm{Im}\varphi$

$$O(n)/SO(n) \cong Z_2 \tag{2.17}$$

- $SU(n) = \{A \in U(n) \mid \det A = 1\}$ (special unitary group) $\forall A \in U(n)$, $\det\big(AA^\dagger\big) = \det A \det A^\dagger = |\det A|^2 = 1$,

$$\Rightarrow \det : U(n) \to U(1) = \{e^{i\varphi} \mid \varphi \in [0,2\pi)\} \tag{2.18}$$

Im det $= e^{i\varphi} \in U(1)$, one can choose $A = \mathrm{diag}(1,1,\ldots,e^{i\varphi})$ thus det is a onto map.

$$\Rightarrow SU(n) \trianglelefteq U(n), \quad U(n)/SU(n) \cong U(1) \tag{2.19}$$

Claim 1

$$GL(n,\mathbb{R}) \cong SL(n,\mathbb{R}) \times \mathbb{R}^* \text{ if } n \text{ is odd.} \tag{2.20}$$

证明

Recall: $G_1, G_2 \trianglelefteq G, G_1 \cap G_2 = \{e\}, G_1G_2 = G$

$$\Rightarrow G \cong G_1 \times G_2 \text{ (Theorem 6)} \tag{2.21}$$

Now, $SL(n,\mathbb{R}) \trianglelefteq GL(n,\mathbb{R})$, $\bar{\mathbb{R}}^* \trianglelefteq GL(n,\mathbb{R})$
where $\bar{\mathbb{R}}^* = \{\lambda\mathbb{I} \in \mathrm{Mat}(n,\mathbb{R}) \mid \lambda \in \mathbb{R}, \lambda \neq 0\} \cong \mathbb{R}^*$
because $\forall A \in GL(n,\mathbb{R})$, $\lambda\mathbb{I} \in \bar{\mathbb{R}}^*$, $A\lambda\mathbb{I} = \lambda\mathbb{I}A$.
If n is odd,

$$SL(n,\mathbb{R}) \cap \bar{\mathbb{R}}^* = \{\mathbb{I}\} \tag{2.22}$$

since $A \in SL(n,\mathbb{R}) \cap \bar{\mathbb{R}}^*$, $\det(A) = \det(\lambda\mathbb{I}) = \lambda^n = 1$

$$\Rightarrow \lambda = 1 \tag{2.23}$$

$SL(n,\mathbb{R}) \cdot \bar{\mathbb{R}}^* = \{A\lambda\mathbb{I} \mid A \in SL(n,\mathbb{R}), \lambda\mathbb{I} \in \bar{\mathbb{R}}^*, n \text{ is odd}\}$.
Suppose $\bar{A} \in GL(n,\mathbb{R})$, $\lambda = (\det \bar{A})^{1/n}$,
thus $A = \frac{1}{\lambda}\bar{A} \in SL(n,\mathbb{R}) : \det A = \frac{1}{\lambda^n} \det \bar{A} = 1$

$$\Rightarrow \bar{A} = \lambda\mathbb{I} \cdot A \Rightarrow SL(n,\mathbb{R}) \cdot \bar{\mathbb{R}}^* = GL(n,\mathbb{R}) \text{ for } n \text{ is odd.} \tag{2.24}$$

Claim 2

$$SU(n) \times U(1)/\mathbb{Z}_n \cong U(n) \text{ compare } U(n)/SU(n) \cong U(1) \tag{2.25}$$

证明

First, we define:

$$\begin{aligned} k : SU(n) \times U(1) &\to U(n) \\ (s, e^{i\varphi}) &\mapsto se^{i\varphi}, \quad \varphi \in [0, 2\pi) \end{aligned} \tag{2.26}$$

we can see that k is a homomorphism:

$$k(s_1, e^{i\varphi_1}) \cdot k(s_2, e^{i\varphi_2}) = k(s_1 s_2, e^{i\varphi_1 + i\varphi_2}) \tag{2.27}$$

$$s_1 e^{i\varphi_1} \cdot s_2 e^{i\varphi_2} = s_1 s_2 e^{i(\varphi_1+\varphi_2)} \tag{2.28}$$

$\ker k \cong \mathbb{Z}_n$:

$$\ker k = \{(s, e^{i\varphi}) \mid se^{i\varphi} = \mathbb{I}, s \in SU(n), \varphi \in [0, 2\pi)\} \tag{2.29}$$

$$\Rightarrow s = e^{-i\varphi}\mathbb{I} \Rightarrow 1 = \det s = \det\left(e^{-i\varphi}\mathbb{I}\right) = e^{-in\varphi} \tag{2.30}$$

$$\Rightarrow n\varphi = 2k\pi \Rightarrow \varphi = \frac{2k\pi}{n}, \quad k = 0, 1, \ldots, n-1 \tag{2.31}$$

$$\Rightarrow s = e^{-i\frac{2k\pi}{n}}\mathbb{I} \tag{2.32}$$

$$\Rightarrow \ker k = \{(e^{-i\frac{2k\pi}{n}}\mathbb{I}, e^{i\frac{2k\pi}{n}}) \mid k = 0, 1, 2, \ldots, n-1\} \tag{2.33}$$

So there is a isomorphism λ

$$\begin{aligned} \lambda : \ker k &\to \mathbb{Z}_n = \{0, 1, \ldots, n-1\} \\ (e^{-i\frac{2k\pi}{n}}\mathbb{I}, e^{i\frac{2k\pi}{n}}) &\mapsto k (\text{mod } n) \end{aligned} \tag{2.34}$$

where the cyclic group can be re-expressed by this form: $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$, where we define the operation is plus. And $1 + 1 + \cdots + 1 = n \equiv 0$, so element 1. e.g. $\mathbb{Z}_3 = \{0, 1, 2\}$,

take any number, say 5, we have $5/3 = 1 \cdots 2 \Rightarrow (5-2)/3 = 1 \Rightarrow 5 \equiv 2(\text{mod } 3)$.
Now we can check λ is indeed a isomorphism.

(a) λ is a homomorphism.

$$\begin{aligned}\lambda(e^{-i\frac{2k_1\pi}{n}}\mathbb{I}, e^{i\frac{2k_1\pi}{n}}) \cdot \lambda(e^{-i\frac{2k_2\pi}{n}}\mathbb{I}, e^{i\frac{2k_2\pi}{n}}) &= k_1 + k_2 + nq, q \in \mathbb{Z} \\ &= \lambda(e^{2i\frac{k_1+k_2}{n}\pi}\mathbb{I}, e^{2i\frac{k_1+k_2}{n}\pi}) = \lambda(s_1 s_2, e^{i\varphi_1}e^{i\varphi_2})\end{aligned} \tag{2.35}$$

(b) λ is one-to-one:

$$\ker\,\lambda = \{(s, e^{i\varphi}) \in \ker\,k \mid \lambda(s, e^{i\varphi}) = 0\} \tag{2.36}$$

where 0 is the neutral element in $\mathbb{Z}_n$

$$\Rightarrow \ker\,\lambda = \{(e^0\mathbb{I}, e^0)\} = \{(\mathbb{I}, 1)\} \tag{2.37}$$

$$\forall k_1 = k_2 \Rightarrow \lambda(e^{-i\frac{2k_1\pi}{n}}\mathbb{I}, e^{i\frac{2k_1\pi}{n}}) = \lambda(e^{-i\frac{2k_2\pi}{n}}\mathbb{I}, e^{i\frac{2k_2\pi}{n}}) \tag{2.38}$$

$$\Rightarrow (s_1, e^{i\varphi_1}) = (s_2, e^{i\varphi_2}) \tag{2.39}$$

(c) λ is onto: trivial.

- k is onto.
  For $k : SU(n) \times U(1) \to U(n)$, $\forall u \in U(n)$

  $$\det u = e^{i\varphi} \text{ with } \varphi \in [0, 2\pi), \tag{2.40}$$

  we can construct

  $$s := e^{-i\frac{\varphi}{n}}u, \quad s^\dagger = u^\dagger e^{i\frac{\varphi}{n}} \Rightarrow ss^\dagger = \mathbb{I}. \tag{2.41}$$

  $$\Rightarrow \det s = e^{-i\varphi}\det u = 1 \Rightarrow s \in SU(n) \tag{2.42}$$

  we then have $(s, e^{i\frac{\varphi}{n}}) \in SU(n) \times U(1)$, thus,

  $$k(s, e^{i\frac{\varphi}{n}}) = se^{i\frac{\varphi}{n}} = u \in U(n) \tag{2.43}$$

Based on the first isomorphism theorem, $SU(n) \times U(1)/\mathbb{Z}_n \cong U(n)$.

Claim 3

$$SL(n,\mathbb{C}) \times \mathbb{C}^*/\mathbb{Z}_n \cong GL(n,\mathbb{C}) \text{ compare } GL(n,\mathbb{C})/SL(n,\mathbb{C}) \cong \mathbb{C}^* \tag{2.44}$$

证明

$k : SL(n,\mathbb{C}) \times \mathbb{C}^* \to GL(n,\mathbb{C})$

$$(A, z) \mapsto zA \tag{2.45}$$

where $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$.

- k is homomorphism:

$$\begin{aligned}k((A_1, z_1)(A_2, z_2)) &= k(A_1A_2, z_1z_2) = (z_1z_2)A_1A_2 \\ &= z_1A_1 \cdot z_2A_2 = k(A_1, z_1) \cdot k(A_2, z_2)\end{aligned} \tag{2.46}$$

- ker $k \cong \mathbb{Z}_n$.

$$\ker\ k = \{(A, z) \in SL(n, \mathbb{C}) \times \mathbb{C}^* \mid k(A, z) = zA = \mathbb{I}\}. \tag{2.47}$$

$$zA = \mathbb{I} \Rightarrow A = z^{-1}\mathbb{I}. \tag{2.48}$$

$$1 = \det A = \det\left(z^{-1}\mathbb{I}\right) = z^{-n} \Rightarrow z = e^{i\frac{2k\pi}{n}}, k = 0, 1, \ldots, n-1 \tag{2.49}$$

$$\Rightarrow A = e^{-i\frac{2k\pi}{n}}\mathbb{I}. \tag{2.50}$$

$$\Rightarrow \ker\ k = \{(e^{-i\frac{2k\pi}{n}}\mathbb{I}, e^{i\frac{2k\pi}{n}}) \mid k \in \{0, 1, \ldots, n-1\}\}. \tag{2.51}$$

$\Rightarrow$ ker $k \cong \mathbb{Z}_n$.

- k is onto: $\forall B \in GL(n, \mathbb{C})$, we can construct: $s = z^{-\frac{1}{n}}\mathbb{I}B$
det : $GL(n, \mathbb{C}) \to \mathbb{C}$, so note: $\det B = ze^{i\varphi}$

$$\Rightarrow \det s = \frac{1}{z}e^{-i\varphi} \det B = 1 \Rightarrow s \in SL(n, \mathbb{C}) \tag{2.52}$$

$\Rightarrow \forall B \in GL(n, \mathbb{C})$, one can find $(s, z^{\frac{1}{n}}e^{i\frac{\varphi}{n}}) \in SL(n, \mathbb{C}) \times \mathbb{C}^*$.

Using the first isomorphism theorem, $SL(n, \mathbb{C}) \times \mathbb{C}^*/\mathbb{Z}_n \cong GL(n, \mathbb{C})$.

Claim 4

$$SL(n, \mathbb{R}) \times \mathbb{R}^*/\mathbb{Z}_2 \cong GL_+(n, \mathbb{R}), \quad n \text{ is even.} \tag{2.53}$$

compare: $GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong \mathbb{R}^*$

证明

Now, we have $GL(n, \mathbb{R}) \cong SL(n, \mathbb{R}) \times \mathbb{R}^*$, for n is odd. This case we consider the even number. Define the map:

$$\begin{aligned} k : SL(n, \mathbb{R}) \times \mathbb{R}^* &\to GL(n, \mathbb{R}) \\ (A, m) &\mapsto m \cdot A \end{aligned} \tag{2.54}$$

- k is homomorphism:

$$\begin{aligned} k((A_1, m_1)(A_2, m_2)) &= k(A_1A_2, m_1m_2) \\ &= (m_1m_2)A_1A_2 = m_1A_1 \cdot m_2A_2 = k(A_1, m_1)k(A_2, m_2) \end{aligned} \tag{2.55}$$

- ker $k = \{(\pm\mathbb{I}, \pm 1)\} \cong Z_2$

$$\ker\ k = \{(A, m) \in SL(n, \mathbb{R}) \times \mathbb{R}^* \mid k(A, m) = mA = \mathbb{I}\} \tag{2.56}$$

$$\Rightarrow A = m^{-1}\mathbb{I}, \quad 1 = \det A = \det\left(m^{-1}\mathbb{I}\right) = m^{-n} \tag{2.57}$$

For n is even, $m = \pm 1$

$$\Rightarrow \ker\ k = \{(\pm\mathbb{I}, \pm 1)\} \cong Z_2 \tag{2.58}$$

Now we want to use $G/\ker\varphi \cong \mathrm{Im}\varphi$. But we find a problem, namely, Im $k = GL_+(n, \mathbb{R})$ for even n. So we need to introduce the following:

- $\operatorname{Im} k = GL_+(n,\mathbb{R})$

$$GL_+(n,\mathbb{R}) = \{B \in GL(n,\mathbb{R}) \mid \det B > 0\} \tag{2.59}$$

which is a subgroup of $GL(n,\mathbb{R})$. For $B \in GL_+(n,\mathbb{R})$, $\det B = z > 0$, one define

$$A = z^{-1/n} B \Rightarrow \det A = z^{-1} \det B = 1. \tag{2.60}$$

Thus, $A \in SL(n,\mathbb{R})$. So, $\forall B \in GL_+(n,\mathbb{R})$, we have

$$k(A, z^{1/n}) = z^{1/n} A = B. \tag{2.61}$$

which proves the map is onto.

Using the first isomorphism theorem,

$$SL(n,\mathbb{R}) \times \mathbb{R}^*/\mathbb{Z}_2 \cong GL_+(n,\mathbb{R}), \quad n \text{ is even}. \tag{2.62}$$